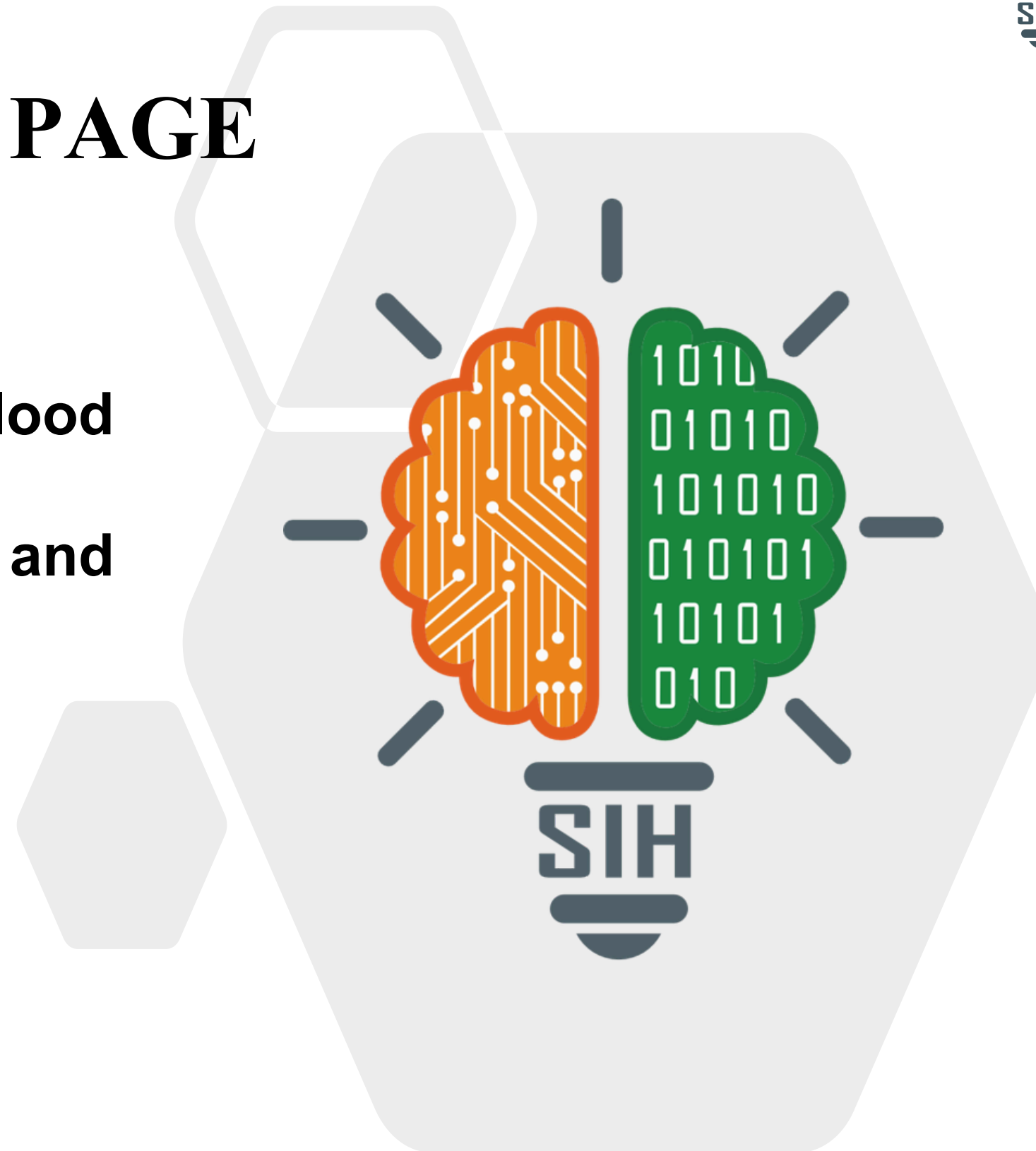


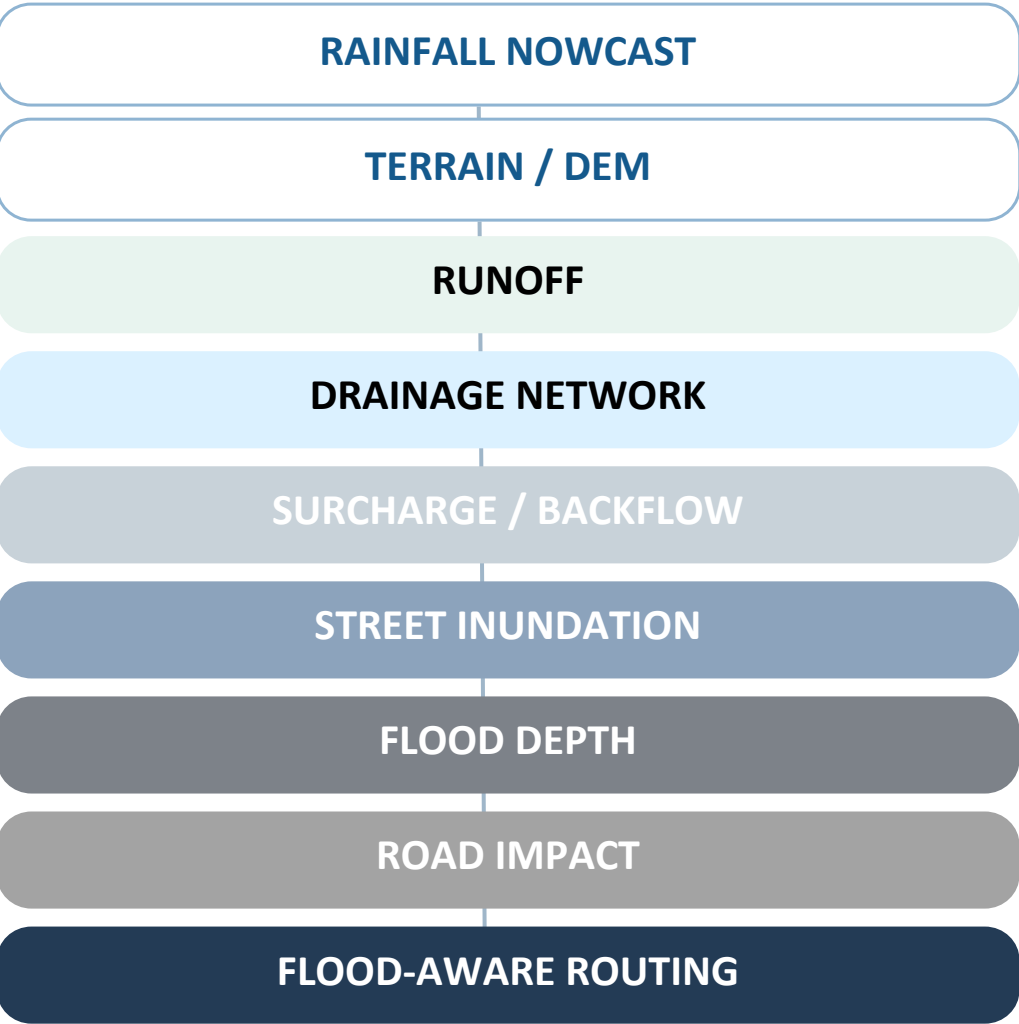
TITLE PAGE

- Problem Statement ID –SIH26085
- Problem Statement Title-Urban Flood
Nowcasting System (Drainage and
Rainfall Coupling)
- Theme-Disaster Management
- PS Category- Software
- Team ID-166925
- Team Name (Singularity@)



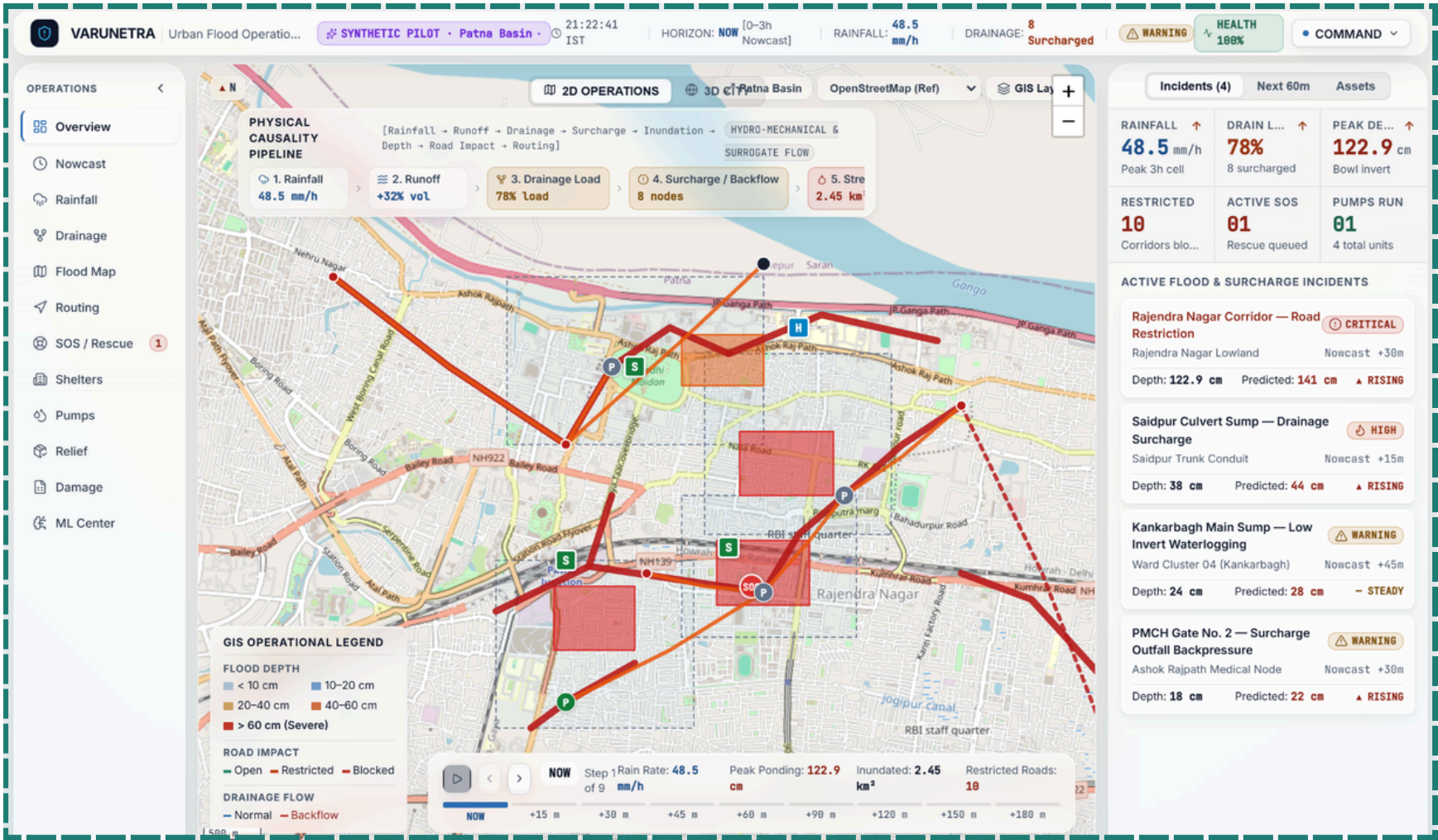
PROBLEM
Rainfall forecast ≠ street-level flood information

HOW IT ADDRESSES THE PROBLEM



PROPOSED SOLUTION

A rainfall–drainage coupled urban flood nowcasting platform that transforms rainfall forecasts into street-level flood-depth, road-impact and actionable response intelligence.

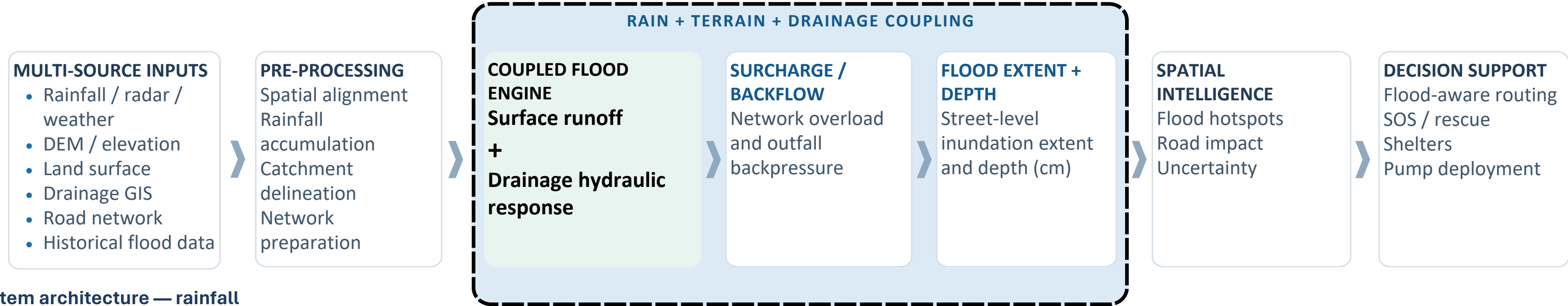


VARUNETRA Main Command Center — 2D Operations
SYNTHETIC PILOT · Patna Urban Basin · simulated outputs, not field-validated

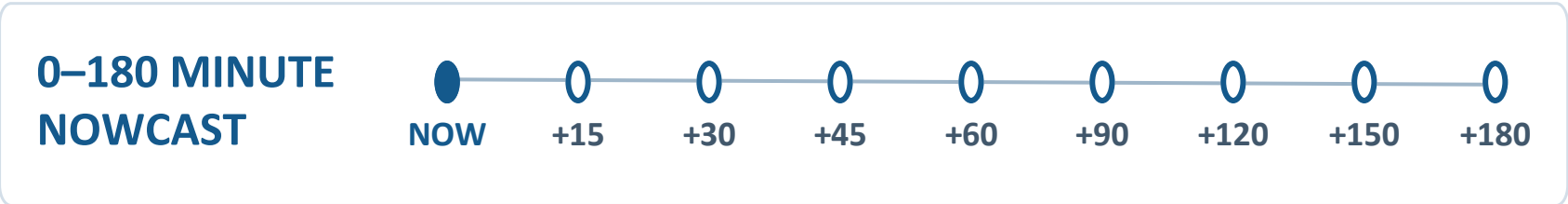
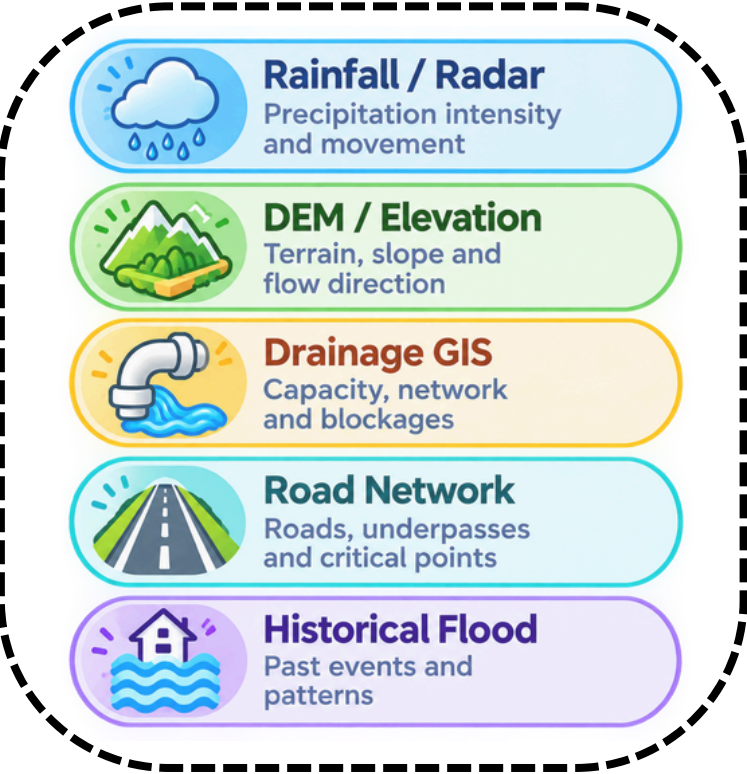
INNOVATION AND UNIQUENESS

- 1. Coupled intelligence**
Rainfall, terrain and drainage are analysed together.
- 2. Street-level output**
Produces spatial flood extent and depth intelligence.
- 3. Drainage-aware modelling**
Considers surcharge and backflow instead of rainfall-only mapping.
- 4. Prediction → Action**
Flood intelligence feeds road-impact analysis and operational routing.

METHODOLOGY AND PROCESS FOR IMPLEMENTATION

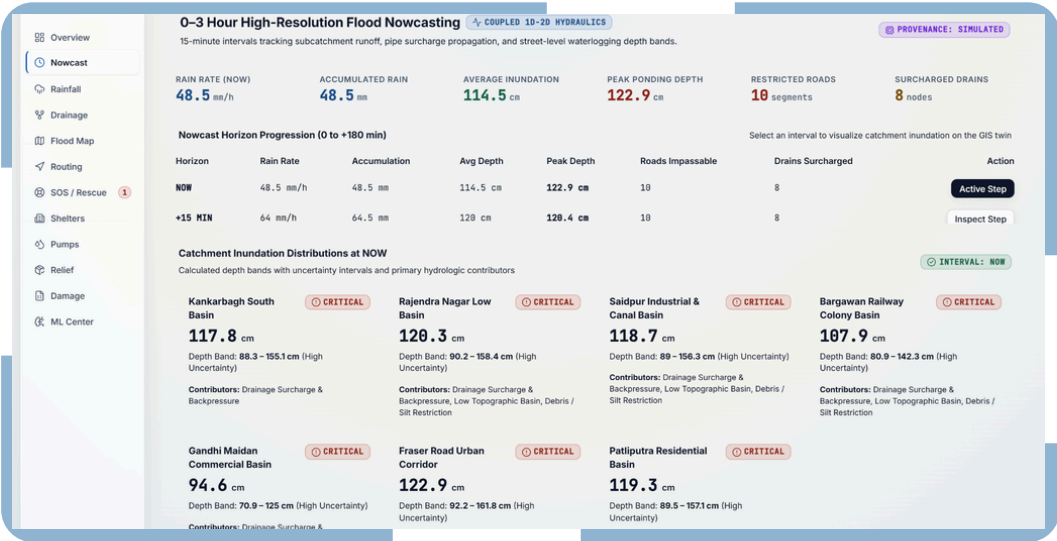


System architecture — rainfall to operational decision support



TECHNOLOGIES TO BE USED

FRONTEND React + TypeScript Leaflet + CesiumJS	BACKEND Python + FastAPI	GEOSPATIAL PostgreSQL + PostGIS	MODELLING Hydrologic / hydraulic simulation Drainage network modelling ML surrogate + explainability
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Prototype evidence • 0-180 minute nowcast

TECHNICAL FEASIBILITY

Pilot geography: synthetic Patna Urban Basin

01
Modular data-provider architecture

02
Coupled surface + drainage modelling

03
GIS-based spatial outputs

04
API-driven decision support

05
Catchment-based deployment

06
2D + 3D visualization

KEY CHALLENGES → MITIGATION

CHALLENGE
Drainage data availability
 Network topology, capacity and asset data may be incomplete.

→

MITIGATION
Data adapters + provenance
 Every layer labelled REAL / SIMULATED / SYNTHETIC / DEMO / CACHED.

CHALLENGE
Local calibration
 Urban hydraulic behaviour varies by catchment.

→

MITIGATION
Event-based calibration
 Use historical rainfall and observed inundation where available.

CHALLENGE
Live-data access
 Operational feeds may require approved credentials.

→

MITIGATION
Physics + ML
 Physics-based modelling as engineering reference; ML as accelerator / surrogate where justified.

CHALLENGE
Computational cost
 High-resolution simulation can be expensive.

→

MITIGATION
Incremental deployment
 Start with a catchment / ward and scale after validation.

PROTOTYPE EVIDENCE
Software / prototype validation only — not field validation.

0–180 min Nowcast

2D GIS

3D City

Routing

SOS

Pumps

11/11 automated backend tests passed

Frontend production build passed



POTENTIAL IMPACT ON TARGET AUDIENCE

MUNICIPAL / DISASTER AUTHORITIES

- Street-level situation awareness
- Drainage stress monitoring
- Flood hotspot identification
- Response prioritisation

EMERGENCY SERVICES

- Flood-aware routing
- SOS prioritisation
- Rescue dispatch support
- Shelter accessibility

CITIZENS

- Localised flood awareness
- Impacted-road visibility
- Alternative-route information
- Shelter discovery

RECOVERY TEAMS

- Damage assessment
- Road restoration prioritisation
- Pump / dewatering coordination
- Post-event analysis

CALCULATED NAVIGATION SOLUTION
SAFEST Profile • LIGHT_VEHICLE

6.8 min
Distance: 4.4 km

OPERATIONAL SAFETY POLICY COMPLIANCE
Restricted by configured operational threshold

ⓘ RESTRICTED

RISK RATING
CRITICAL

MAX DEPTH
104 cm

RECALCULATION
0.024s

⚠ AVOIDED INUNDATION HAZARDS (4)

Kankarbagh Main Road: Predicted depth 104.0 cm exceeds configured operational policy threshold (22 cm) for LIGHT_VEHICLE.

Rajendra Nagar Overbridge Approach: Predicted depth 100.0 cm exceeds configured operational policy threshold (22 cm) for LIGHT_VEHICLE.

Saidpur Canal Road: Predicted depth 98.0 cm exceeds configured operational policy threshold (22 cm) for LIGHT_VEHICLE.

Bypass Link to Patna Junction: Predicted depth 73.6 cm exceeds configured operational policy threshold (22 cm) for LIGHT_VEHICLE.

Routing console • synthetic pilot

Policy-based clearance check — decision support, not a safety guarantee.

BENEFITS OF THE SOLUTION

SOCIAL

Earlier and more localised flood awareness.

OPERATIONAL

Better prioritisation of roads, drainage nodes and emergency resources.

ECONOMIC

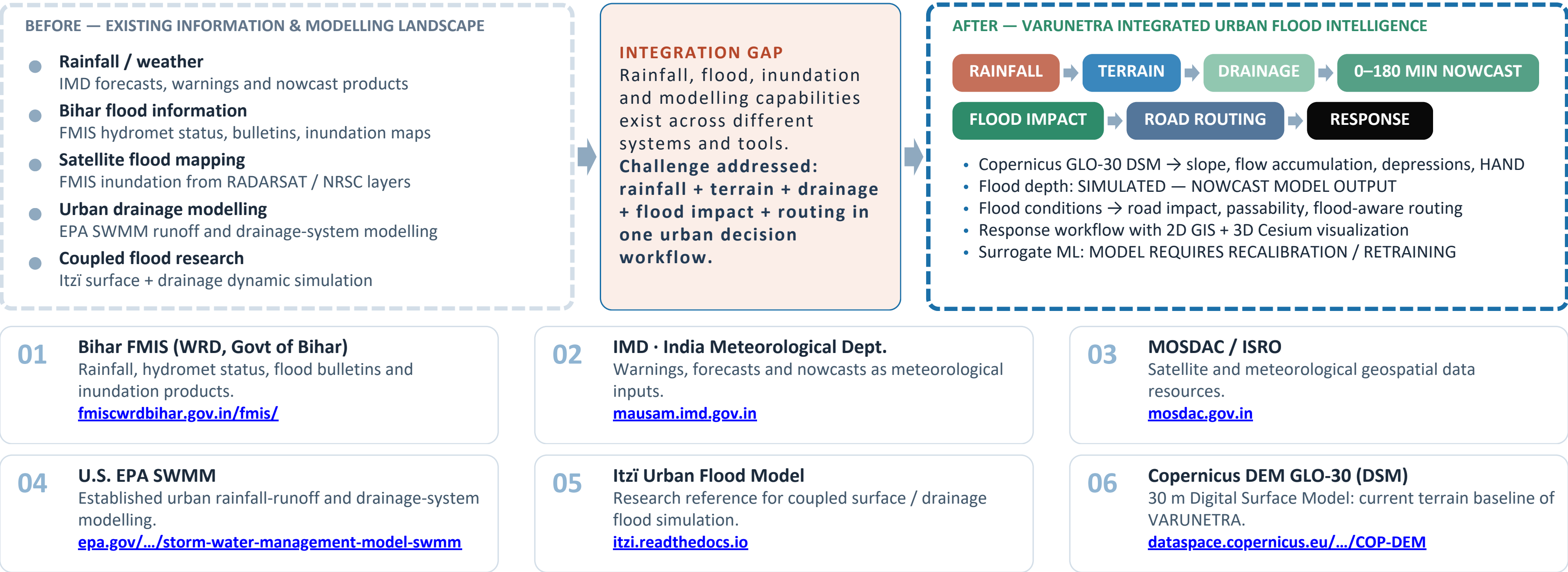
Potentially reduced disruption through earlier operational decisions.

ENVIRONMENTAL

Supports targeted drainage and dewatering interventions.

BEFORE → AFTER • EVIDENCE-BASED EVOLUTION

From Distributed Flood Information to Integrated Urban Flood Intelligence



DATA PROVENANCE

REAL INPUT

DERIVED OUTPUT

SIMULATED / ESTIMATED

DEMO CONTROL

VARUNETRA is a team-developed prototype. It is not an official government product. References are provided as methodological, data-source and ecosystem references.